HYDRAULIC LASH ADJUSTER (HLA) REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

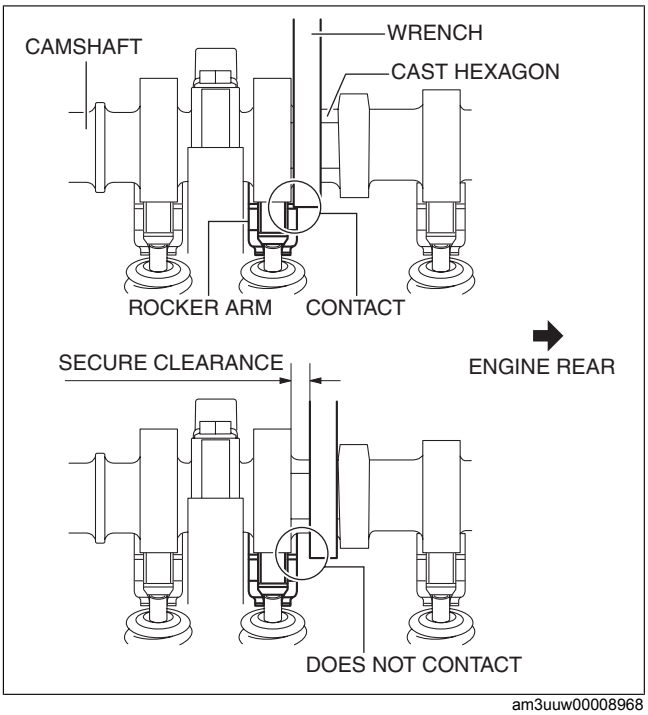
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Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

Caution

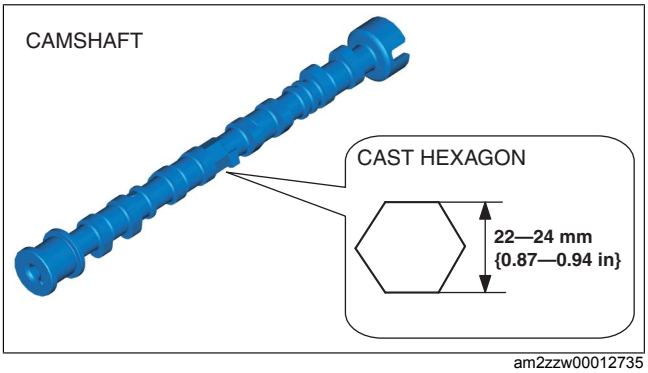
- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**



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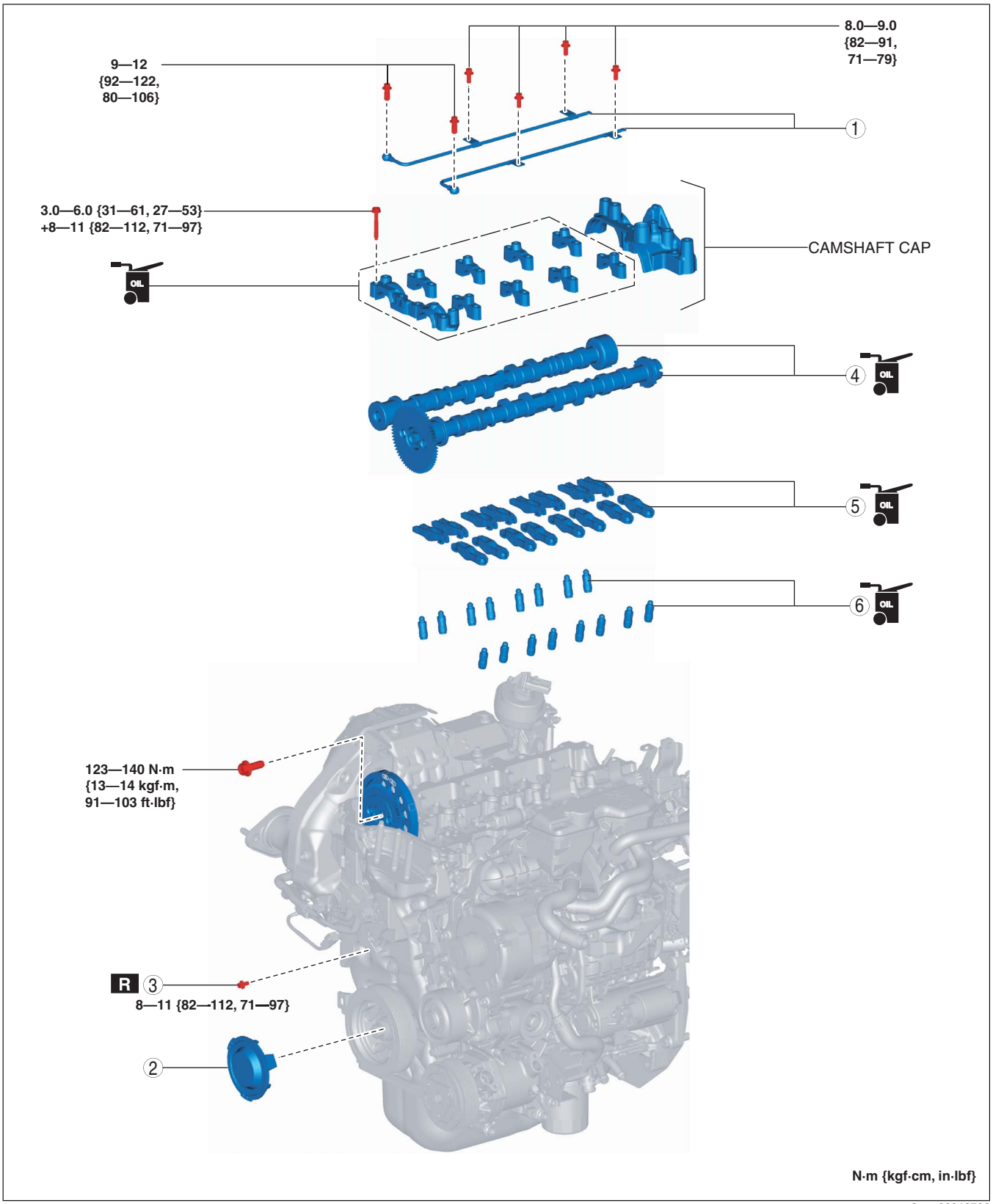
Note

- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}.**



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1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the PCM and PCM bracket as a single unit. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Remove the drive belt auto tensioner. (See DRIVE BELT AUTO TENSIONER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Remove the fuel injectors. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Remove the lower case. (See LOWER CASE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7. Remove the vacuum pump. (See VACUUM PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8. Remove the supply pump. (See SUPPLY PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9. Remove the cylinder head cover. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
10. Remove the CMP sensor. (See CAMSHAFT POSITION (CMP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
11. Remove the front splash shield. (RH) (See SPLASH SHIELD REMOVAL/INSTALLATION.)
12. Remove in the order indicated in the table.
13. Install in the reverse order of removal.
14. Start the engine and inspect the following:
    - Idle speed (See ENGINE TUNE-UP [SKYACTIV-D 1.5].)



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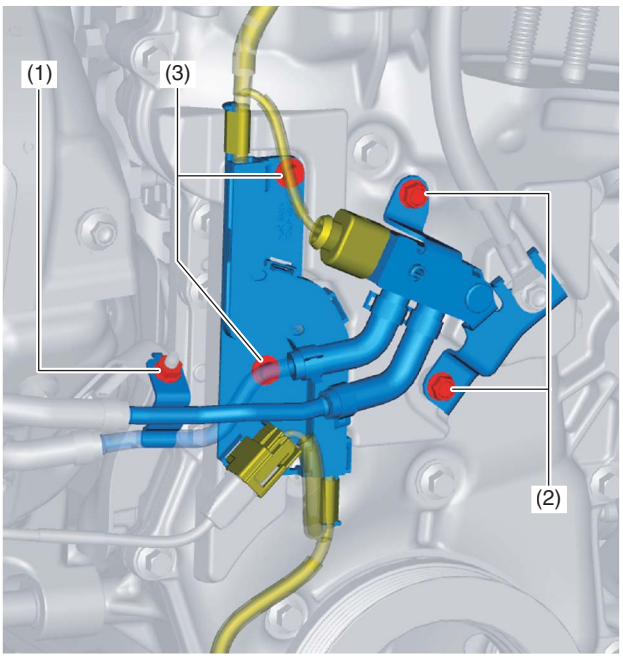
1	Oil shower pipe (See Oil Shower Pipe Installation Note.)
2	Crankshaft pulley cover (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].)
3	Blind plug (See Blind Plug Removal Note.) (See Blind Plug Installation Note.)

4	Camshaft (See Camshaft Removal Note.) (See Camshaft Installation Note.)
5	Rocker arm (See Rocker Arm Removal Note.) (See Rocker Arm Installation Note.)

6	HLA (See HLA Removal Note.) (See HLA Installation Note.)

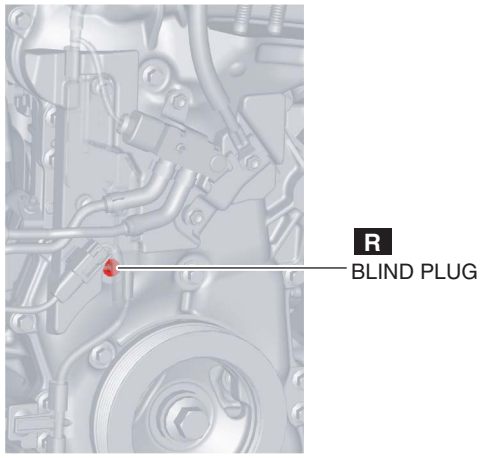
Blind Plug Removal Note

1. Remove the following parts in the order of the numbers shown in the figure.
   (1) Air pipe stay installation nut
   (2) Exhaust gas pressure sensor No.2 bracket installation bolts
   (3) Wiring harness protector installation bolts



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2. Remove the blind plug on the engine front cover service hole.



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Camshaft Removal Note

Caution

- **Keep the timing chain and camshaft sprocket pulled up during and after the work in which it has been set aside from the exhaust camshaft. If the chain falls down, it may disengage from the sprocket on the crankshaft side and deviate, leading to valve timing deviation.**
- **After servicing, always verify the valve timing.**

1. Release the tension on the timing chain using the following procedure:

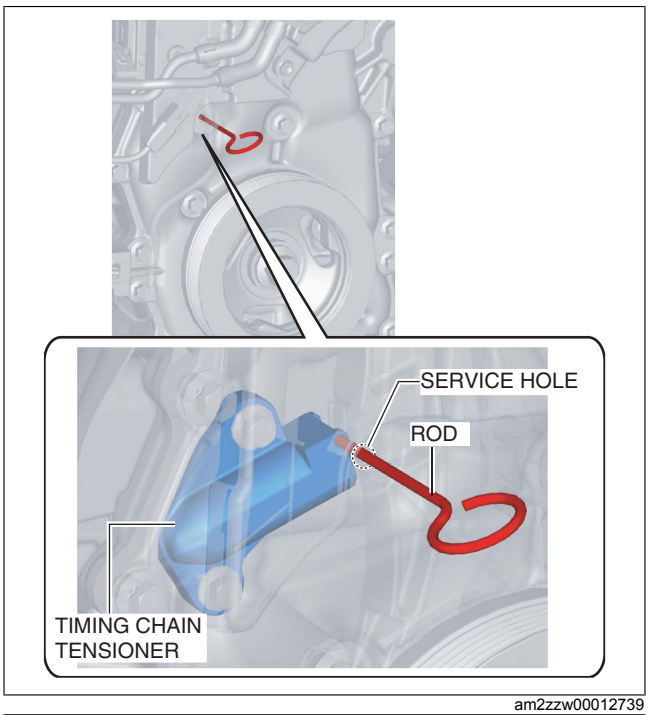
Note

- Release the tension on the timing chain by securing the plunger of the timing chain tensioner through the service hole on the engine front cover.

(1) Insert a metal rod **(diameter approx. 3.2 mm {0.13 in}, length approx. 60—70 mm {2.4 — 2.7in})** into the hole in the body of the timing chain tensioner through the service hole.

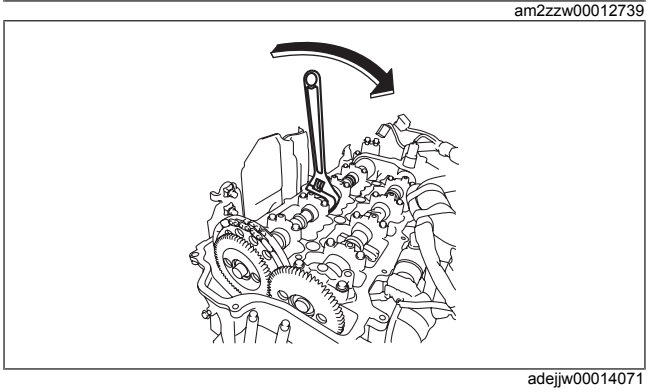
Note

- Before inserting the timing chain tensioner plunger, insert the rod **approx. 22 mm {0.87 in}** from the edge surface of the engine front cover.



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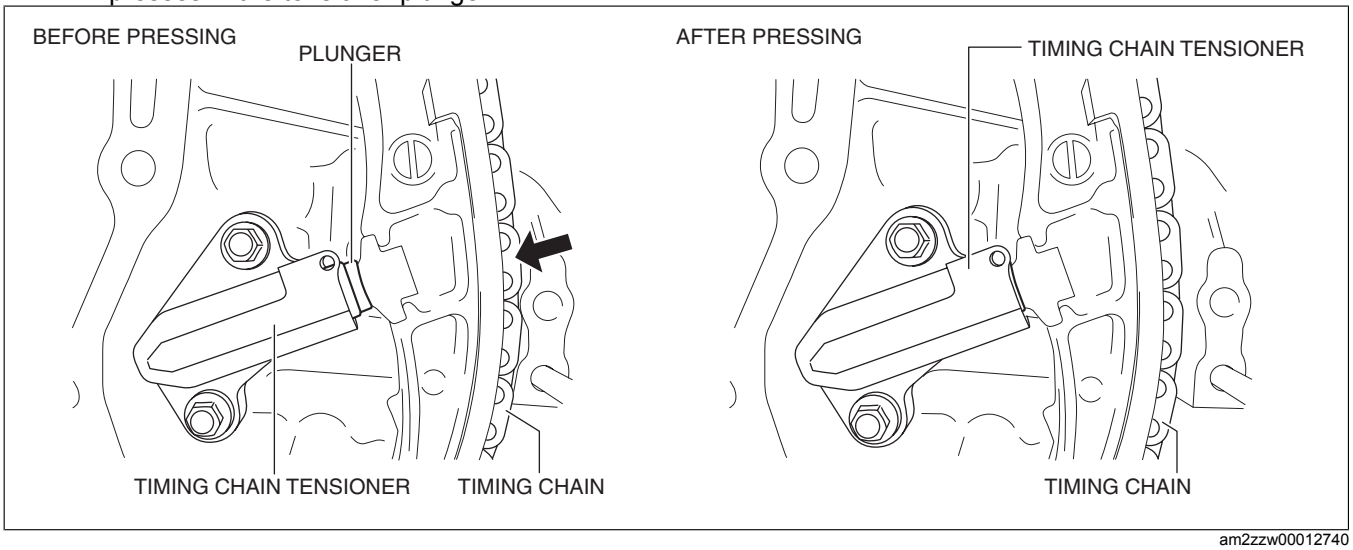
(2) While moving the exhaust camshaft in the direction of the arrow using a wrench on the cast hexagon, press on the plunger of the timing chain tensioner.



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Note

- By rotating the exhaust camshaft clockwise, the timing chain can be pulled slightly and the timing chain presses in the tensioner plunger.



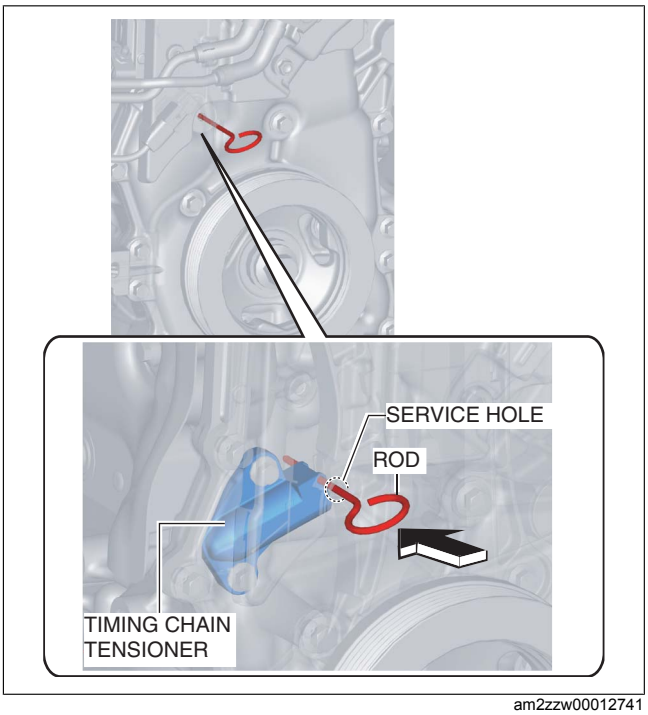
am2zzw00012740

(3) Press on the rod set in Step 1 further with the plunger pressed.

Note
- With the timing chain tensioner plunger inserted, insert the rod a further **approx. 28 mm {1.1 in}**.
- The rod secures the plunger and releases the tension.

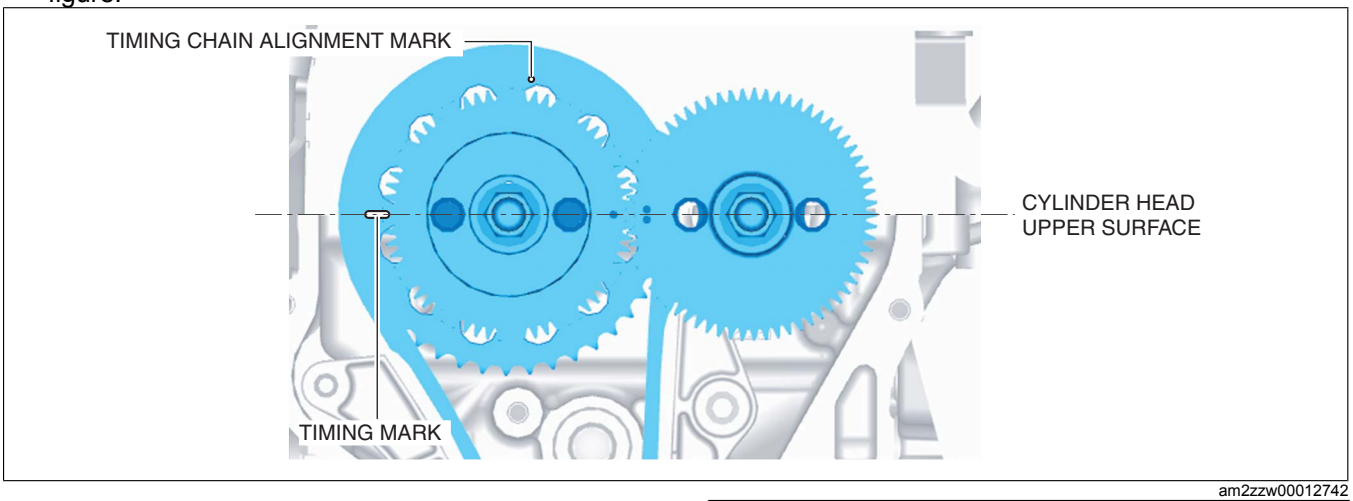
(4) Loosen your grip on the exhaust camshaft and verify that the pressed-in rod does not move.

Note
- If the rod moves easily, the tension on the chain has not been released. Select an appropriate-sized rod and perform the procedure (1) to (4) again.



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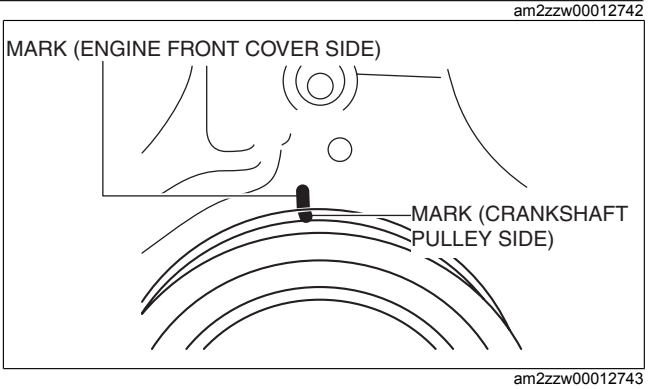
2. Rotate the crankshaft clockwise so that cylinder No.1 is positioned at top dead center (TDC) as shown in the figure.



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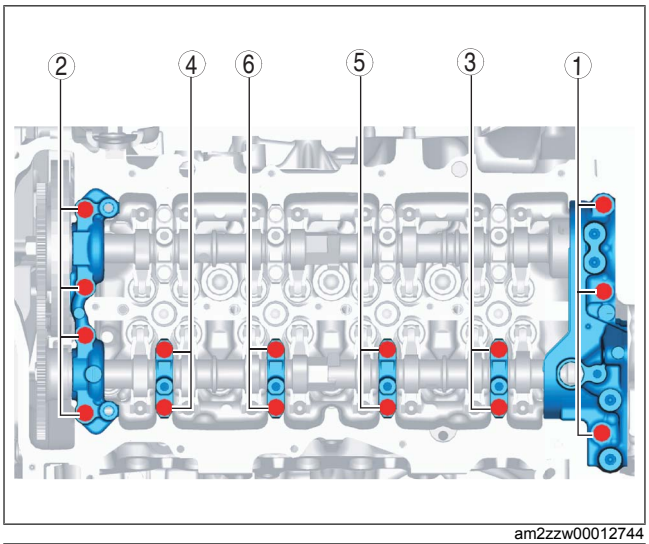
3. Mark the engine front cover and crankshaft pulley so that the valve timing can be verified when assembling.

Note
- The marked position is fine as long as visual verification is possible after assembly.



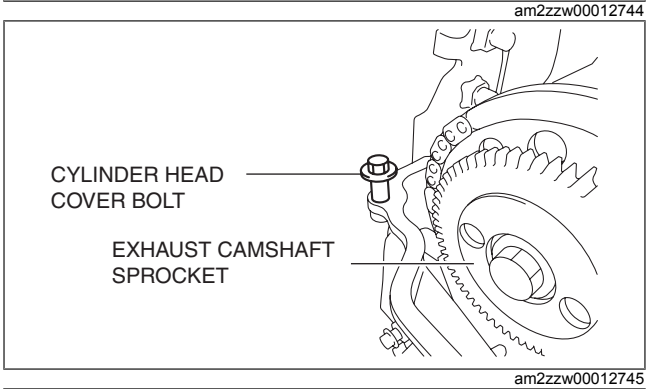
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4. Loosen the intake camshaft cap installation bolts in a few passes in the order shown in the figure and remove the intake camshaft caps.
5. Remove the intake camshaft.



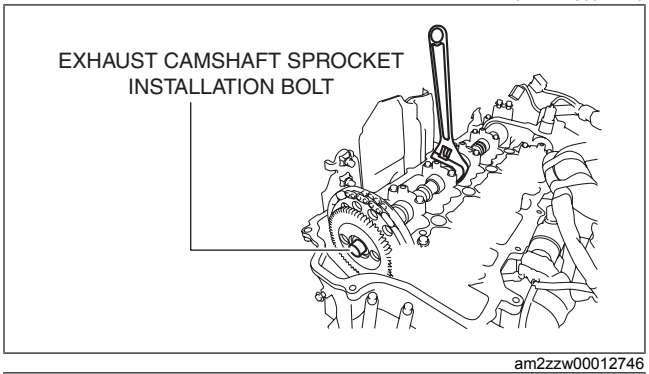
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6. Temporarily install the cylinder head cover bolt to the position shown in the figure to use it for suspending the timing chain and camshaft sprocket.



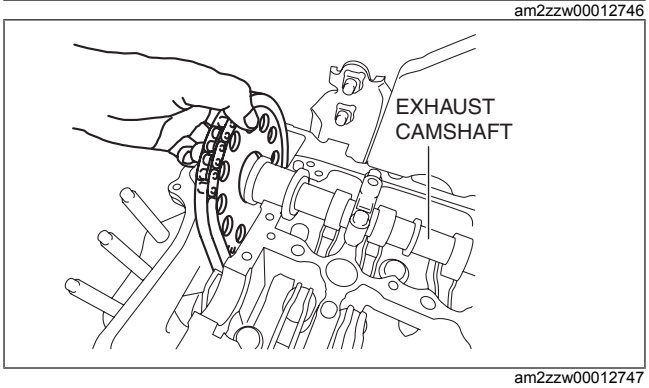
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7. Hold the exhaust camshaft using a wrench on the cast hexagon and remove the exhaust camshaft sprocket installation bolt.



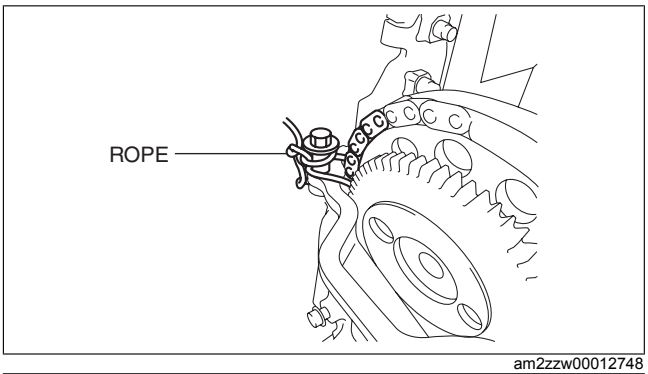
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8. Separate the camshaft sprocket and the exhaust camshaft while the timing chain is hung on the camshaft sprocket.



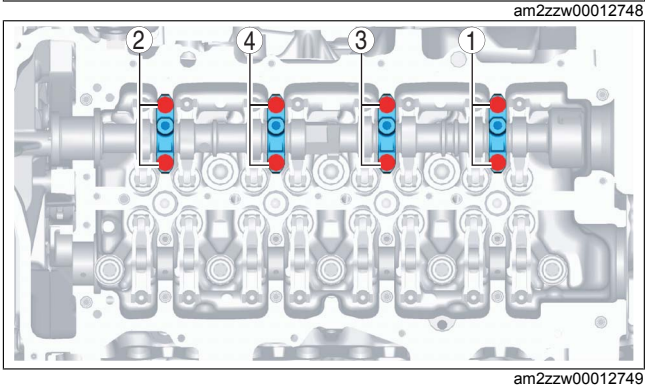
am2zzw00012747

9. Pull up the camshaft sprocket using a rope while the timing chain is hung on the camshaft sprocket and pulled it up as shown in the figure.



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10. Loosen the exhaust camshaft cap installation bolts in a few passes in the order shown in the figure and remove the exhaust camshaft caps.
11. Remove the exhaust camshaft.



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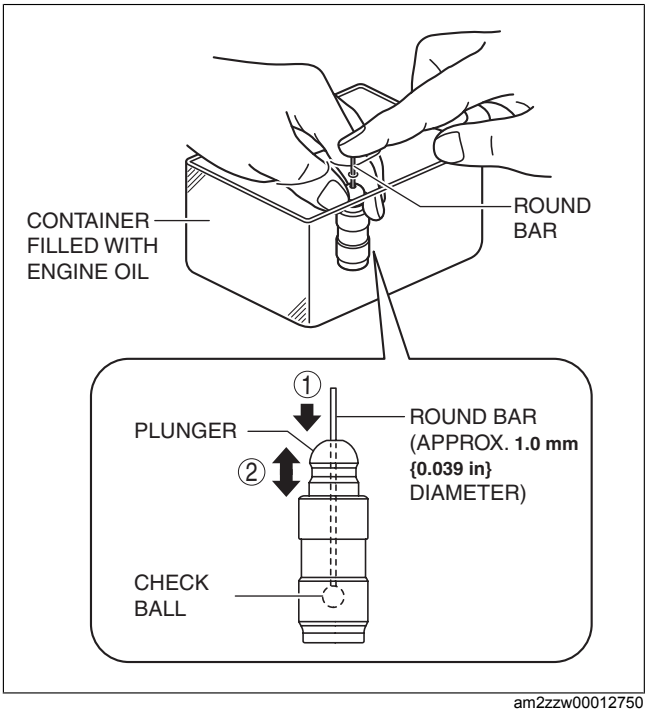
Rocker Arm Removal Note
1. Keep the rocker arms in the order of removal to enable reassembly in their original positions.

HLA Removal Note
1. Keep the HLAs in the order of removal to enable reassembly in their original positions.

HLA Installation Note
1. Perform HLA air bleeding using the following procedure.
   (1) Put the HLA in a container filled with engine oil.

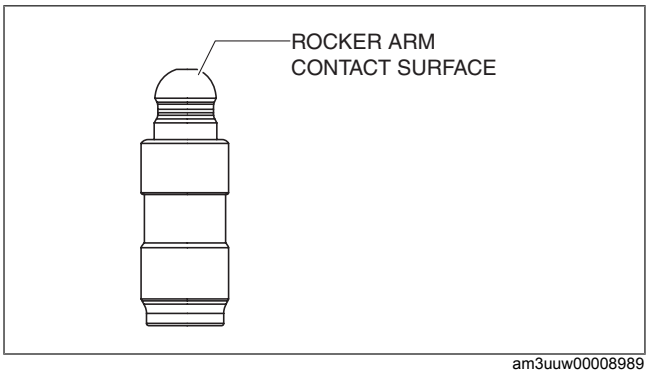
Caution
- **Do not insert the round bar firmly because the check ball spring force is extremely weak.**

   (2) While lightly pressing the check ball using a round bar (**approx. 1.0 mm {0.039 in}** diameter), bleed air by moving the plunger up and down.
   (3) Press the end of the plunger in the oil and verify that there is no rebounding feel.
      - If rebounding feel cannot be eliminated, replace the HLA.



am2zzw00012750

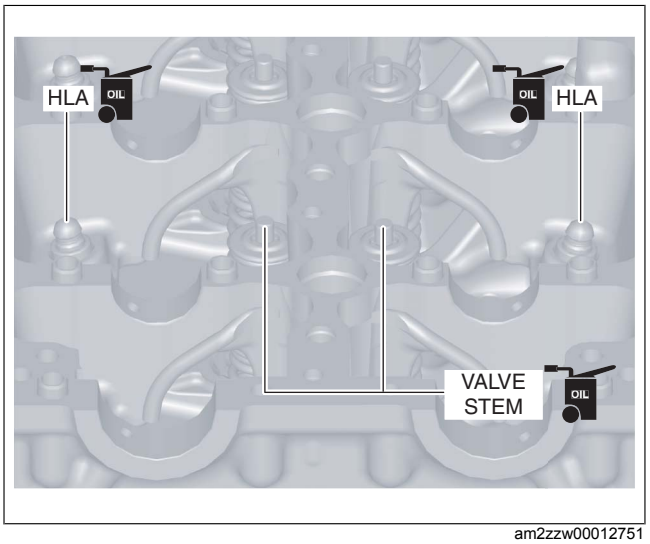
2. Visually inspect the HLA surface where it contacts the rocker arm for wear or damage.
   - If there is any malfunction, replace the HLA.
3. Assemble the HLAs to their original positions.



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Rocker Arm Installation Note

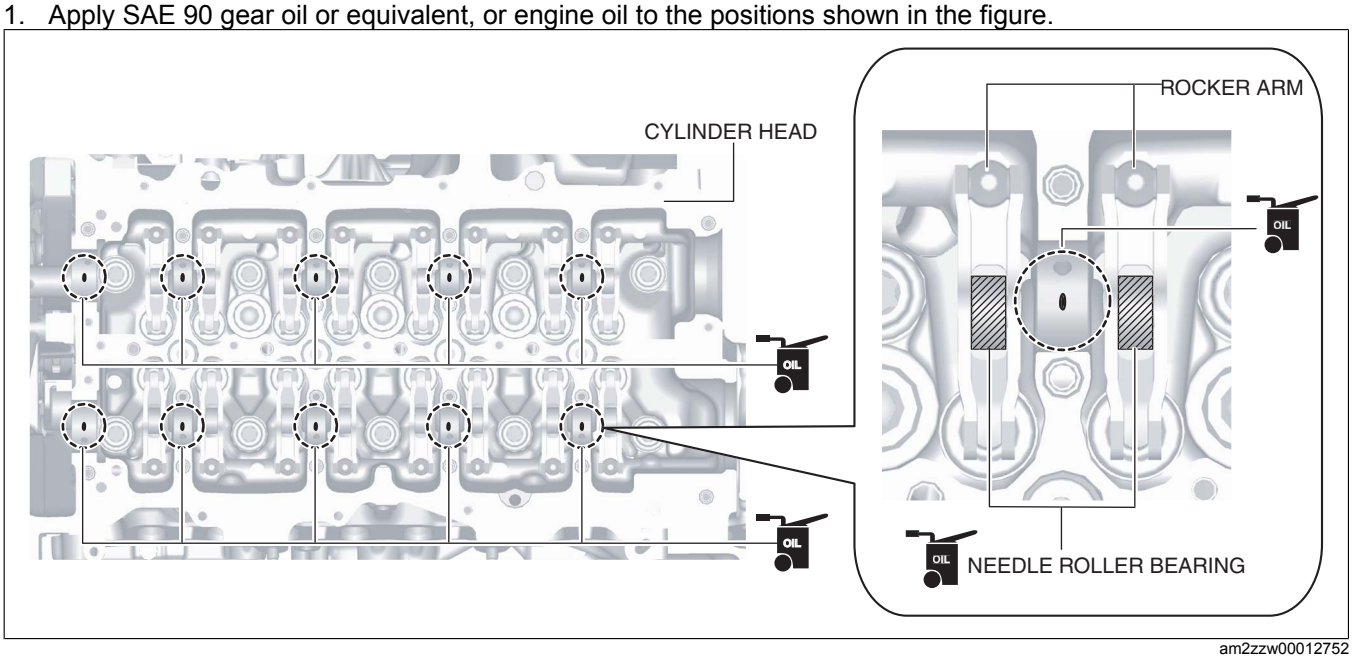
1. Apply engine oil to the HLAs and the end of the valve stems.
2. Install the rocker arms to the same positions as before removal.



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Camshaft Installation Note

1. Apply SAE 90 gear oil or equivalent, or engine oil to the positions shown in the figure.



am2zzw00012752

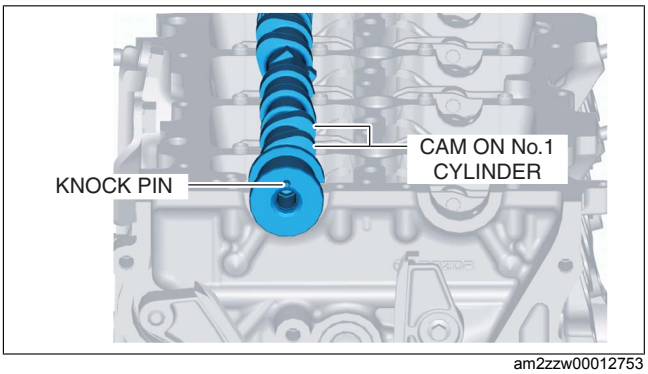
2. Apply SAE 90 gear oil or equivalent, or engine oil to the following locations of each camshaft.
   - Gear sliding surfaces
   - Thrust surface of front journal (both surfaces front and back)

Note

- If oil is applied to the front camshaft cap, oil should not be applied to the thrust surface of the front journal.

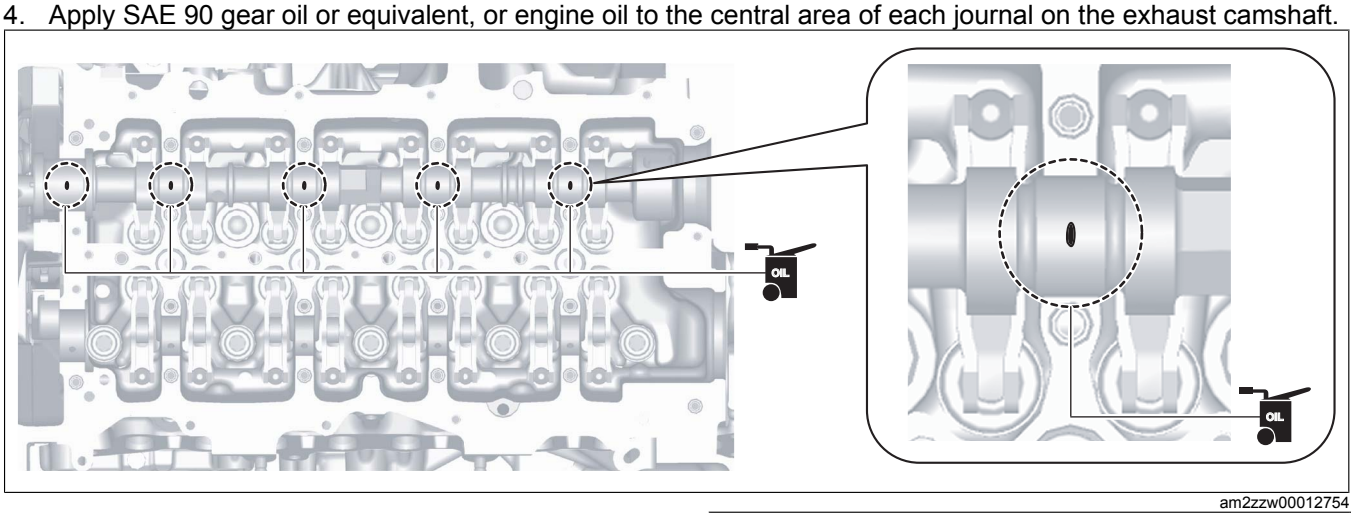
3. Set the exhaust camshaft on the cylinder head.

Note
- By placing the cams on the No.1 cylinder in the position shown in the figure, the notch on the exhaust camshaft sprocket will be positioned close to the knock pin of the exhaust camshaft which will facilitate assembly of the sprocket.



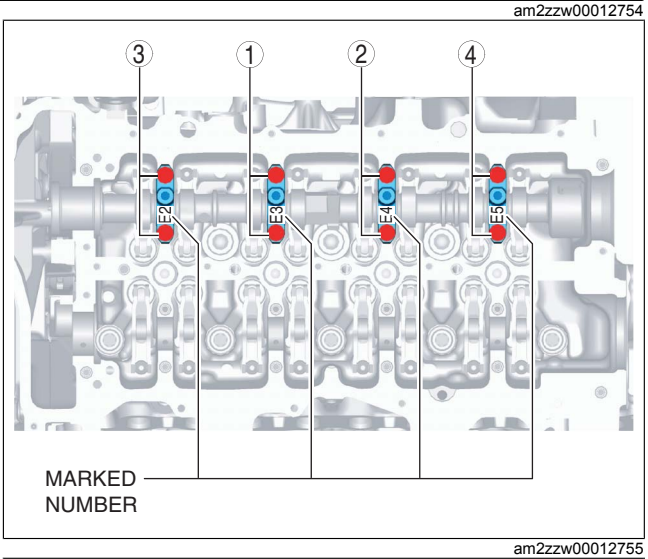
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4. Apply SAE 90 gear oil or equivalent, or engine oil to the central area of each journal on the exhaust camshaft.



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5. Install the exhaust camshaft caps (marked numbers: E2 to E5) in the order of the marked numbers and temporarily tighten the exhaust camshaft cap installation bolts evenly in **2—3 rounds**.
6. Pull the timing chain and the camshaft sprocket and remove the suspending rope.

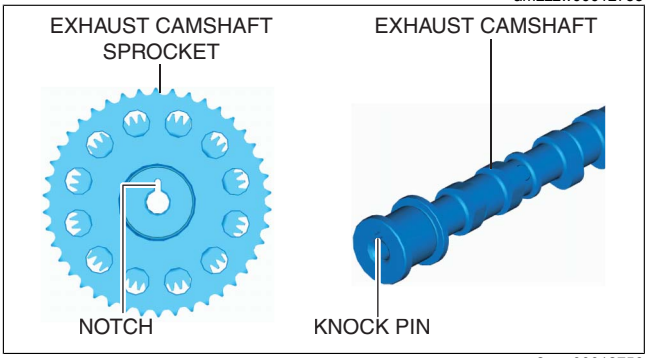


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7. Align the knock pin on the end of the exhaust camshaft with the notch on the camshaft sprocket side, then assemble the sprocket.

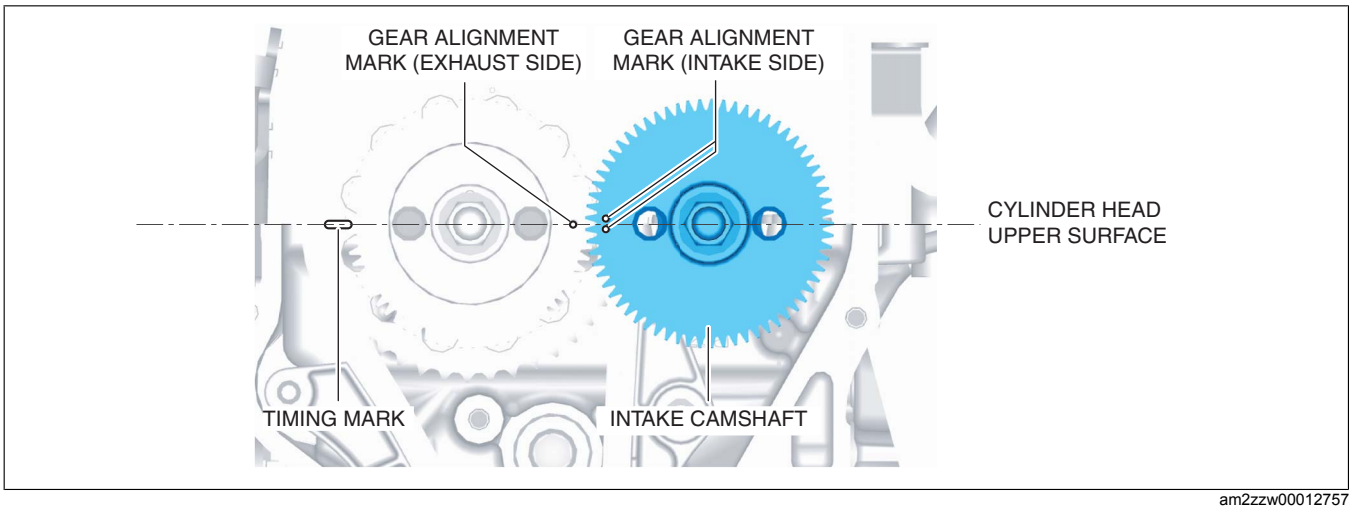
Note
- Align the knock pin and the notch while adjusting the knock pin position by rotating the camshaft.

8. Temporarily tighten the camshaft sprocket installation bolts until they are seated.



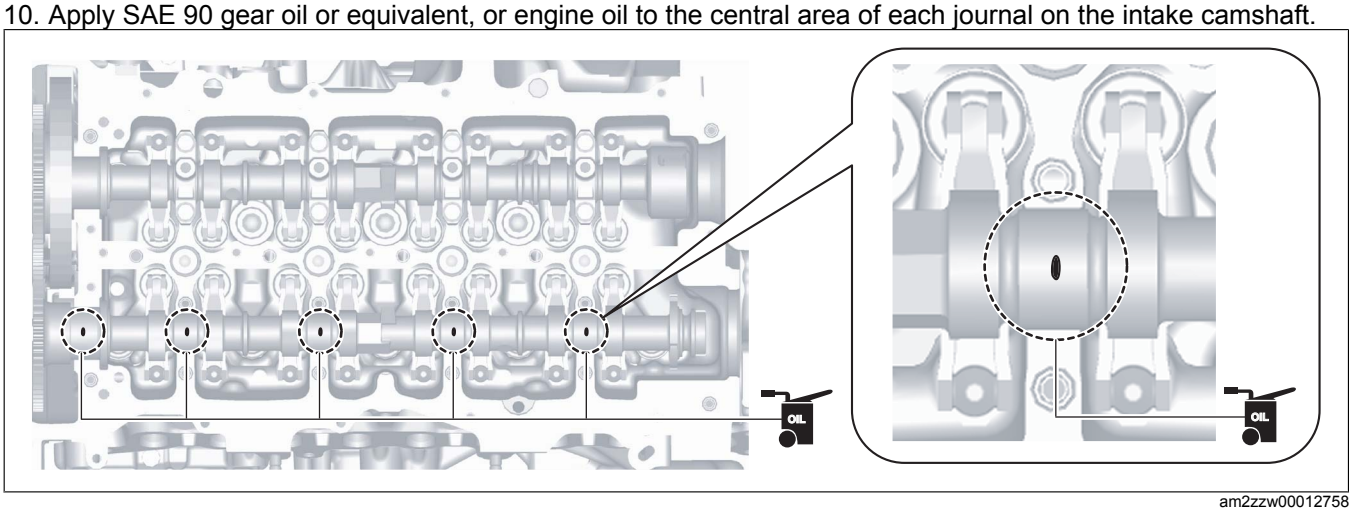
am2zzw00012756

9. Place the intake camshaft on the cylinder head and align the alignment marks for each gear as shown in the figure.



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10. Apply SAE 90 gear oil or equivalent, or engine oil to the central area of each journal on the intake camshaft.



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11. Apply SAE 90 gear oil or equivalent, or engine oil to the thrust surface of the front camshaft cap.

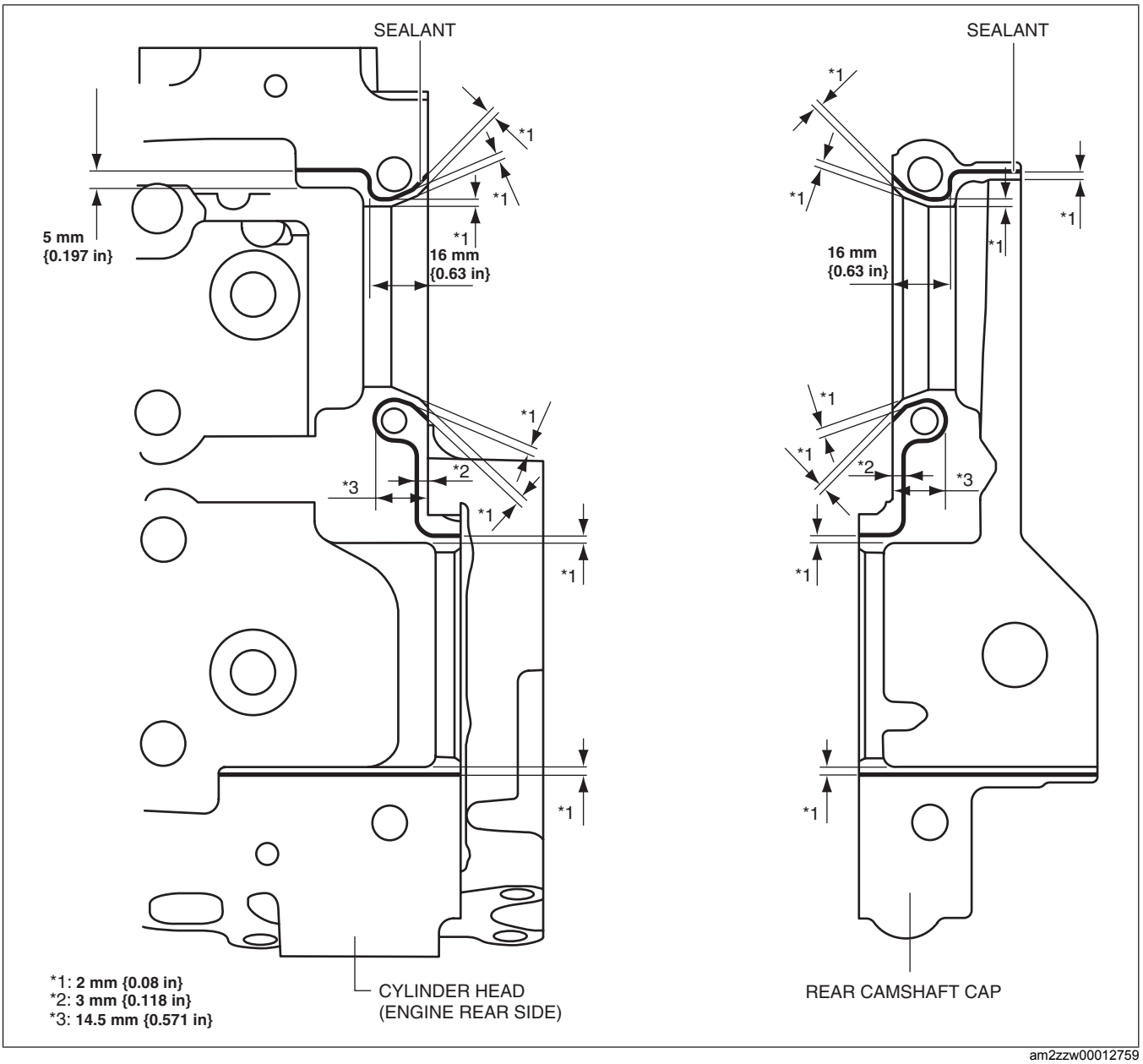
Note

- If oil is applied to the front journal thrust surface of each camshaft, oil should not be applied to the front camshaft cap.

12. Apply sealant (Loctite#962T or equivalent) to the rear side of the cylinder head or rear camshaft cap.

Note

- To prevent engine oil leakage, seal the journal by applying sealant to the rear side of the cylinder head or rear camshaft cap.



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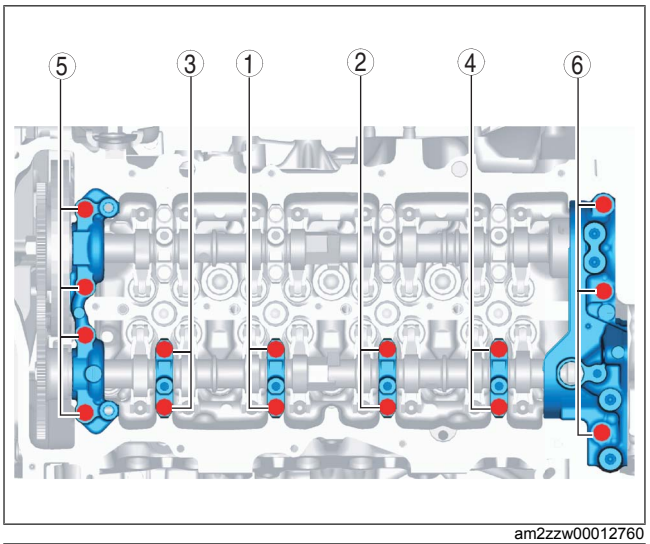
Caution

- **Do not spill sealant on the journal.**
- **To prevent sealant from hardening, adhere the rear camshaft cap to the cylinder head within 10 min after sealant is applied. Tighten the installation bolts completely soon after adhering.**

Sealant bead width

1—3 mm {0.04—0.11 in}

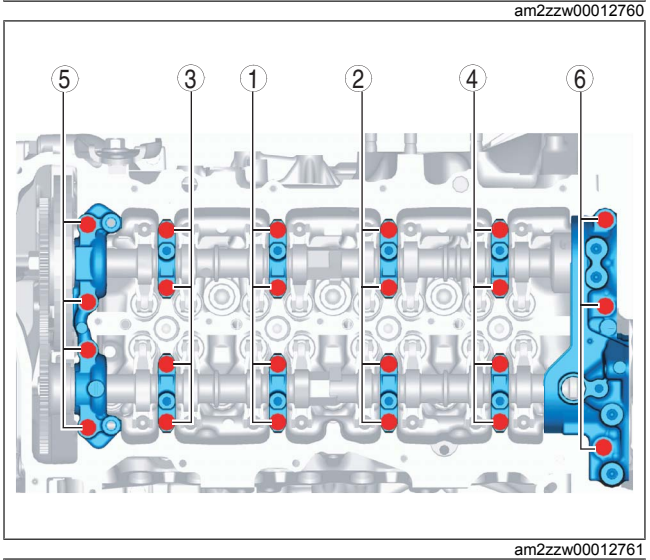
13. Install the camshaft caps (front and rear marked numbers: I2 to I5) in the order of the marked numbers and temporarily tighten the camshaft cap installation bolts evenly in **2—3 rounds**.



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14. Tighten the camshaft cap installation bolts in two steps in the order shown in the figure.

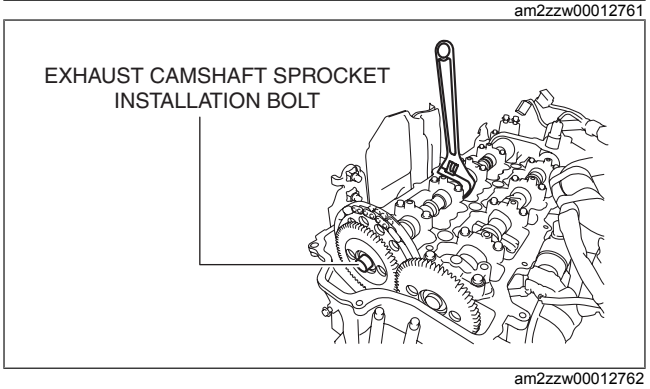
    Tightening torque
    Step 1: 3—6 N·m {31—61 kgf·cm, 27—53 in·lbf}
    Step 2: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



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15. Hold the exhaust camshaft using a wrench on the cast hexagon, and tighten the exhaust camshaft sprocket installation bolt.

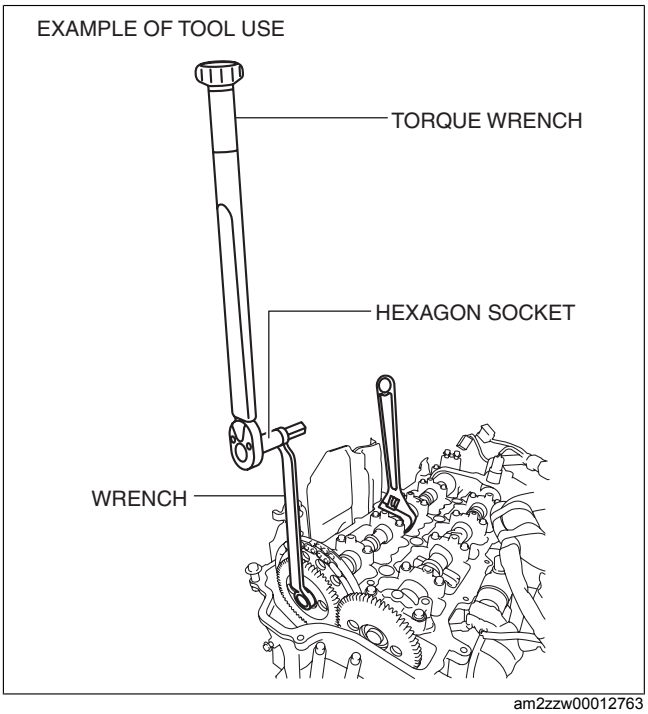
    Tightening torque
    123—140 N·m {13—14 kgf·m, 91—103 ft·lbf}



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Note

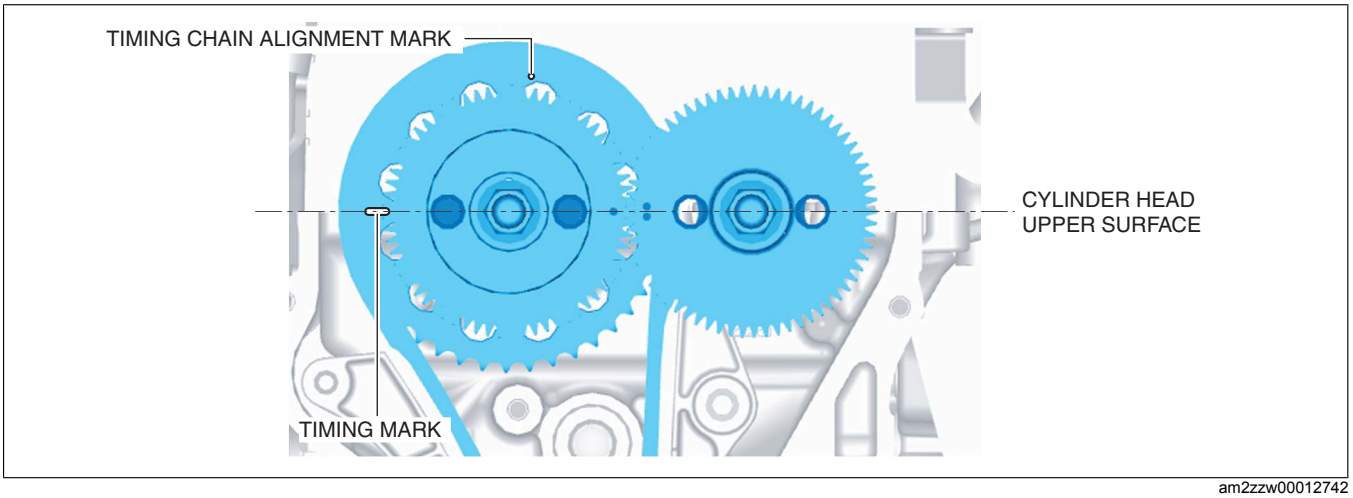
- If there is no appropriate tool which can be used to tighten the camshaft sprocket installation bolt to the specified tightening torque, tighten the bolt using the method as shown in the figure. For the specified tightening torque calculation method if the tool is elongated, refer to the “SERVICE CAUTIONS” in the general information. (See SERVICE CAUTIONS.)



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16. Remove the rod securing the plunger of the timing chain tensioner and apply tension to the timing chain.
17. Rotate the crankshaft clockwise **twice** and verify that the valve timing is correct.

Camshaft side

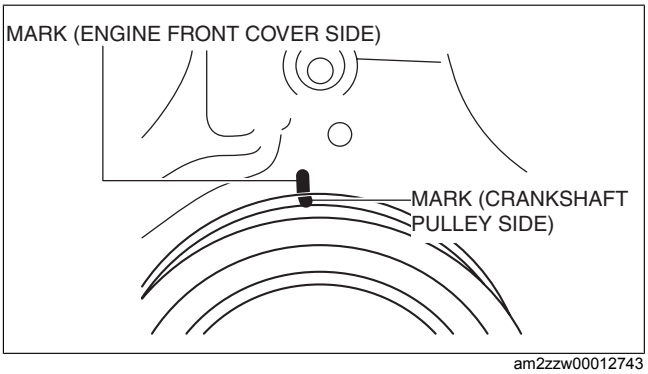


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Crankshaft side

- If the valve timing is not correct, remove the engine front cover and reassemble the timing chain to each sprocket. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

18. Remove the temporarily assembled cylinder head cover bolt.

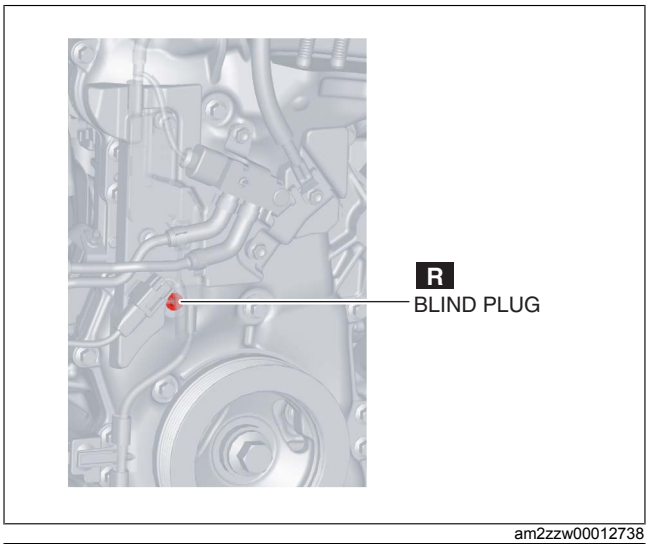


am2zzw00012743

Blind Plug Installation Note

1. Install the new blind plug to the engine front cover service hole.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



am2zzw00012738

2. Install the following parts in the order of the numbers shown in the figure.
   (1) Wiring harness protector installation bolts

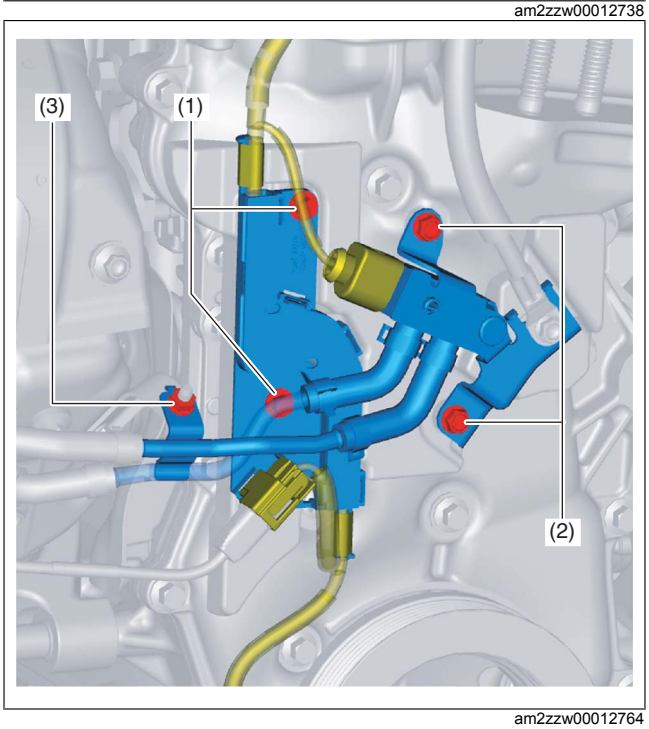
   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

   (2) Exhaust gas pressure sensor No.2 bracket installation bolts

   Tightening torque
   8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}

   (3) Air pipe stay installation nut

   Tightening torque
   8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}



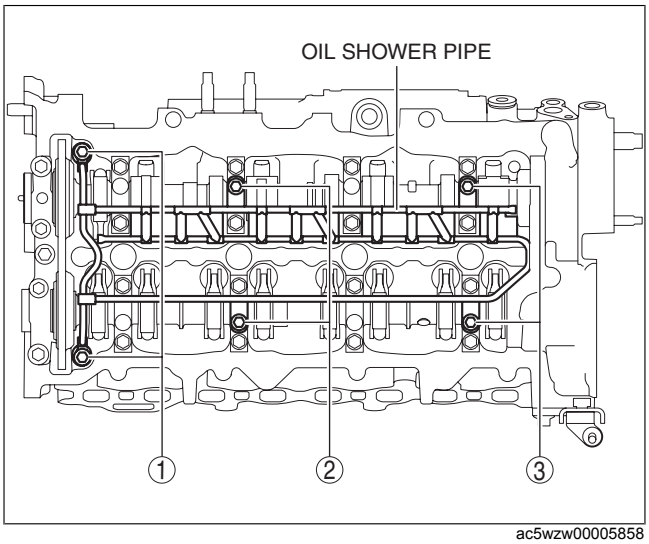
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Oil Shower Pipe Installation Note

1. Install the oil shower pipe in the order shown in the figure.

Tightening torque

Installation position	Tightening torque
1	9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}
2, 3	8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf}



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